

NEERAJ PREMKUMAR

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OBJECTIVE

Driven by curiosity and precision, with a PG Diploma in VLSI Design and Verification and hands-on expertise in RTL design, CMOS fundamentals, and simulation workflows. Proficient in Cadence Virtuoso and industry-standard verification tools, aiming to contribute to advanced semiconductor design and verification projects.

EDUCATION

International PG Diploma in VLSI Design and Verification

Jun 2024- May 2025

IBIS Academy, Thrissur (Accredited by IACET CEUs)

CGPA: 2.63/4.00

Relevant Coursework: CMOS Design, AMBA, RTL & SoC Design Verification.

B.Tech in Electronics and Communication Engineering

Sep 2020- Jun 2024

APJ Abdul Kalam Technological University

CGPA: 6.41/10

Vidya Academy of Science and Technology, Thrissur

Relevant Coursework: Digital Electronics, VLSI Design, System Architecture, Semiconductor Devices

TECHNICAL SKILLS

Languages & Methodologies: Verilog, SystemVerilog, UVM

Verification Techniques: SVA, Constrained Random Verification

Design Concepts: Digital Electronics, FSM, RTL Design

EDA Tools: Cadence Virtuoso, Synopsys VCS, Mentor Graphics QuestaSim, Synopsys Design Compiler

Protocols: UART, AMBA - APB, AHB, AXI

Programming Languages: C, Python (with Data Structures & Algorithms)

PROJECTS

AHB to APB Bridge Verification

Jan 2025 - Feb 2025

- Developed APB master and slave modules in Verilog with support for read/write transactions and proper handshaking.
- Simulated protocol behavior to validate timing control and functional correctness.
- Strengthened understanding of AMBA protocols and verification strategies through waveform analysis.

1x3 Router Design and Verification

Oct 2024 – Nov 2024

- Designed a 1x3 router in Verilog to manage data routing across multiple output channels with efficient control logic.
- Verified uninterrupted data flow and latency reduction through simulation and waveform analysis.
- Utilized ModelSim, EDA Playground, and Vivado for testing, debugging, and performance validation.

Real-Time Vehicle Tracking and Fuel Monitoring System

Sep 2023 - May 2024

- Integrated GPS and fuel sensors for real-time data acquisition, transmitting information to Firebase cloud.
- Created a Kodular app that retrieves and displays live vehicle location and fuel levels via Firebase.

CERTIFICATIONS

- **Introduction to Design Verification using UVM** - IEEE CASS SBC, TKMCE (Mar 2025)
Covered UVM methodology and testbench creation for functional verification
- **Python Programming Training** - Spoken Tutorial Project, IIT Bombay (Aug 2023)
Focused on scripting fundamentals and automation.
- **Arduino Programming Workshop** - NIT Calicut (Nov 2022)
Introduced embedded programming and sensor interfacing.